

Project Report

Fire & Smoke Detection via Image

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Group No: 07

Course Name: CSE468

Section: 01

1. Project Introduction

Fire and smoke detection is a critical area of research within the field of computer vision and intelligent surveillance systems. Traditional fire detection systems rely primarily on physical sensors such as heat or smoke detectors, which are often limited in their response time and spatial coverage. In contrast, image-based detection methods leverage the power of machine learning and deep learning algorithms to identify fire and smoke patterns in real time through visual data captured by cameras.

This project aims to develop an automated fire and smoke detection system using image processing techniques. By training a model on a carefully curated dataset of fire and smoke images, the system will be capable of detecting the presence of fire or smoke in a given image with high accuracy. Such a system has wide-ranging practical applications, including in smart building management, forest fire early warning systems, and industrial safety monitoring.

The project was proposed and initiated by our group following its formation on 2 March. We were allocated a period of two weeks to complete the initial phases, which included project selection, repository setup, and dataset collection.

2. Week 1 — Project Proposal & Selection

During the first week, the group convened to discuss and evaluate a range of potential project topics. After thorough deliberation, we collectively agreed to pursue Fire and Smoke Detection via Image as our project, given its technical relevance and real-world impact.

Key Activities & Individual Contributions:

- The group was officially formed on 2 March.
- Multiple project ideas were proposed and assessed based on feasibility, scope, and relevance.
- The project on Fire and Smoke Detection via Image was selected unanimously.
- Initial roles and responsibilities were assigned among group members.
- Following the proposal selection, I took the initiative to research existing literature and prior work related to fire and smoke detection systems.
- I reviewed several image classification and object detection approaches to understand the most suitable methodology for our project.
- I prepared a brief summary document outlining the proposed system architecture and expected outcomes, which was shared with the team for feedback and alignment.

3. Week 2 — Research & Project Justification

During the second week, the team focused on validating and strengthening the decision to proceed with the chosen project. While other members worked on the technical setup, my primary contribution this week was conducting in-depth research to confirm that the Fire and Smoke Detection via Image project was well-suited in terms of scope, feasibility, and available resources.

Key Activities & Individual Contributions:

- A shared Git repository was created by the team to facilitate collaborative development and version control.
- Git Link: <https://github.com/Starner22/Cse468-Fire-and-Smoke-Detection>
- The team identified and began collecting a dataset of fire and smoke images for model training.
- I conducted an extensive review of existing fire and smoke detection systems, exploring deep learning architectures such as CNNs and YOLO-based models to determine the most effective approach for our use case.
- I investigated publicly available datasets and assessed their quality, size, and relevance to ensure suitability for training a reliable model.
- I compiled a research summary and presented my recommendations to the group, helping the team make informed decisions regarding the technical direction of the project.